



Sequential Logic – Counter Circuit

Logic Design of Digital Systems (300-1209) section 1

LECTURE 10

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Outline

- ☐ Review Flip-Flop
- ☐ Asynchronous Counter (วงจรรนับไม่เข้าจังหวะ)
 - ☐ Ripple Counter
 - ☐ Ripple Counter (JK Flip-Flop)
 - ☐ Ripple Counter (T Flip-Flop)
 - ☐ Modulo-M Counter
- ☒ Synchronous Counter (วงจรรนับเข้าจังหวะ)
- ☐ Variable Modulus Counter
- ☐ Self Stopping Counter (ปกติวงจรรนับจะวนไปเรื่อยๆ)



Review: Flip-Flop

Review: Flip-Flop

① RS Flip-Flop

- NOR Gate
- NAND Gate

$R = \text{Reset}, S = \text{Set}$

Truth Table:

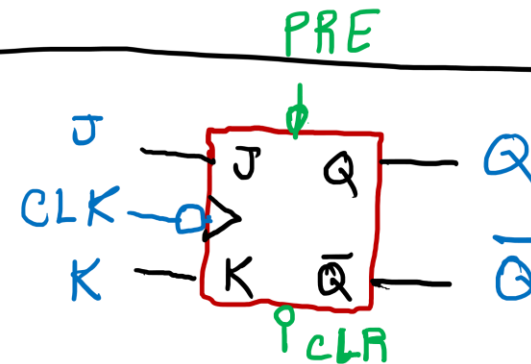
R, S	Q
$(0, 0)$	$Q = \text{ค่าเดิม}$
$(0, 1)$	$Q = 1$
$(1, 0)$	$Q = 0$
$(1, 1)$	$Q = \text{หาไม่ได้}$

② JK Flip-Flop → ကန့်သတ် RS (မာသ်ခဲခဲ)
 $J = \text{set}$
 $K = \text{Reset}$

J	K	Q
0	0	သေသေ
1	0	1
0	1	0
1	1	မာသ်ခဲခဲ

③ D Flip-Flop $\rightarrow$ Delay
(Simple)

④ T Flip-Flop $\rightarrow$ Toggle



Counter Circuit

Counting circuits is an application of the flip-flop for counting numbers in binary system.

- ❑ The number of flip-flops used is equal to the number bits.
- ❑ In the given of the binary number circuit The number of countable values is called Modulus (Mod).

Type of Counter Circuit

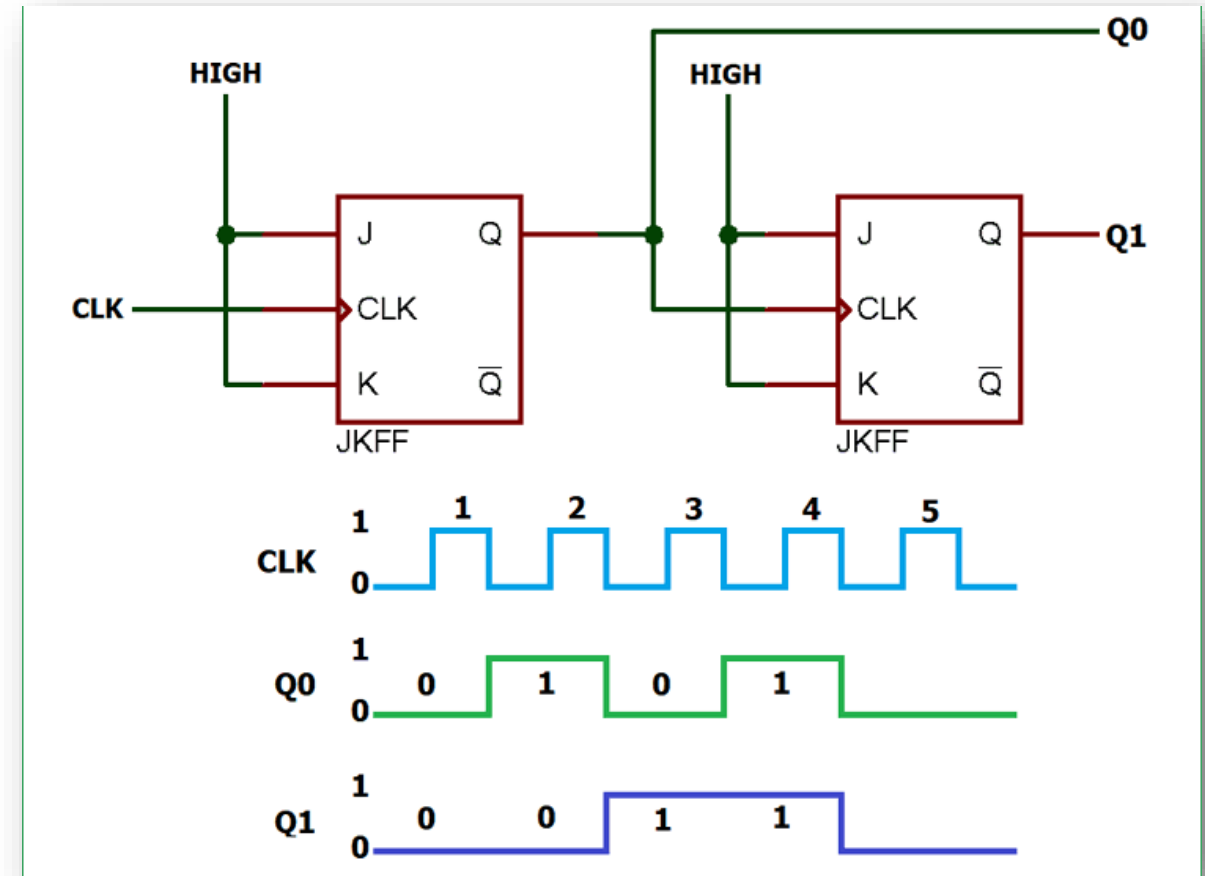
1. Asynchronous Counter
2. Synchronous Counter

1. Asynchronous Counter

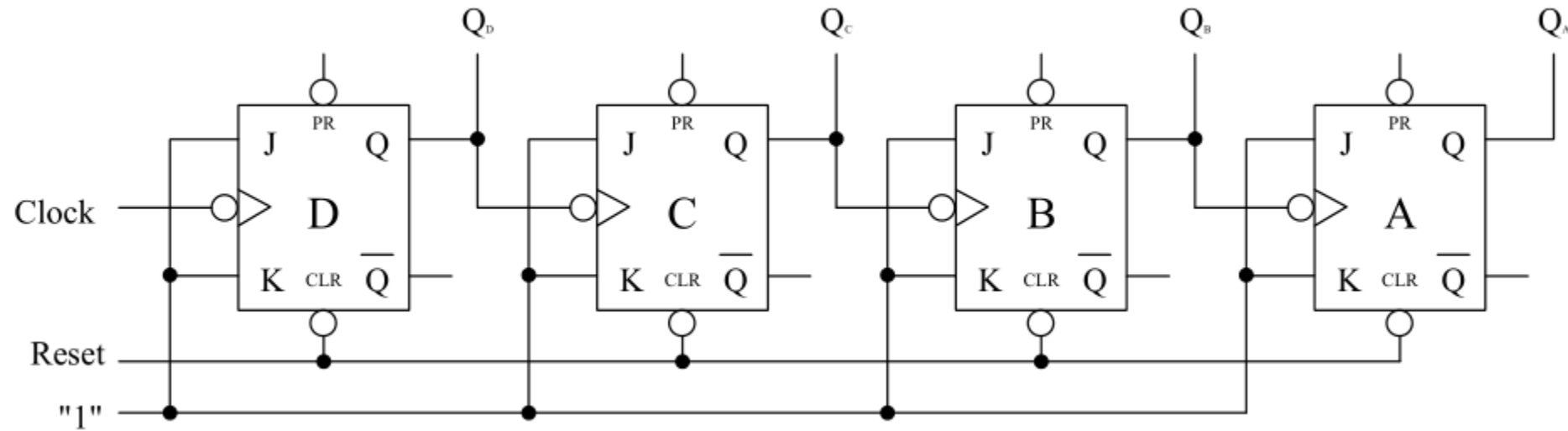
Each flip-flop does not work at the same time. (Clock pulse signal connect only first flip-flop)

[A] Ripple Counter

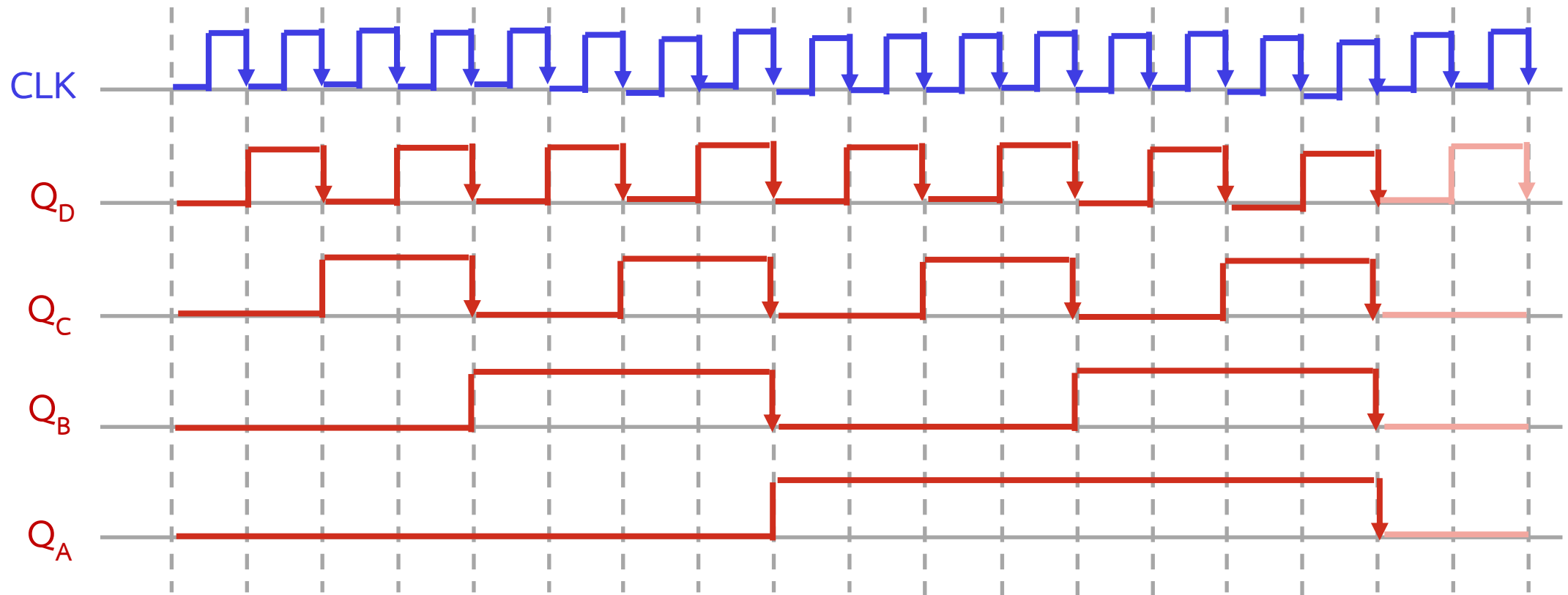
[B] Modulo-M Counter



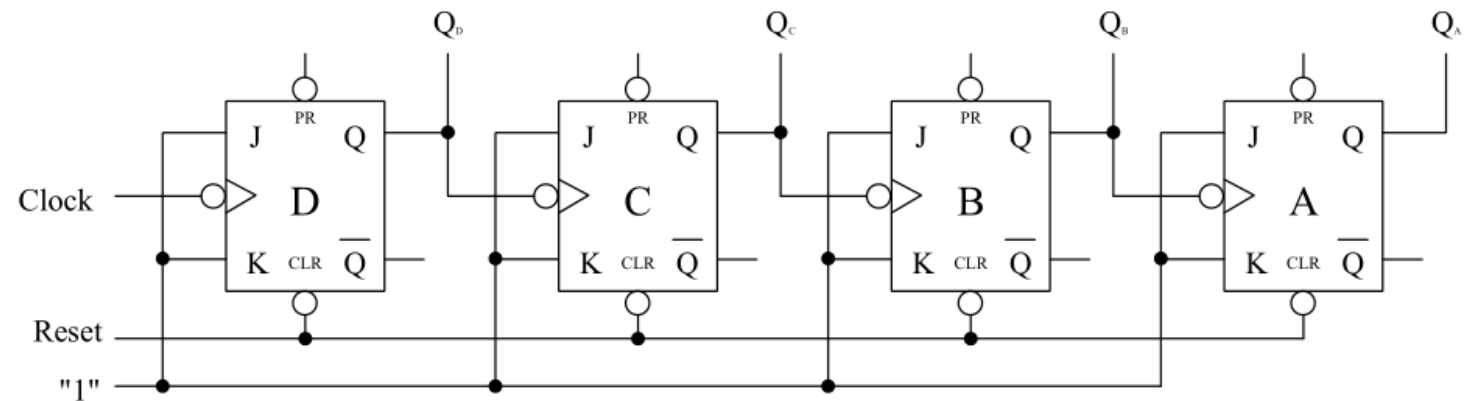
1. [A] Ripple Counter



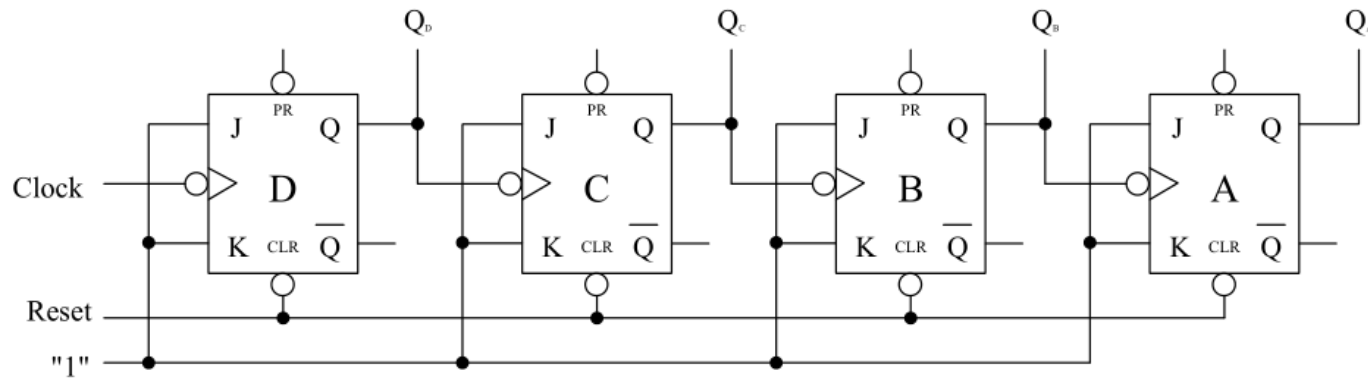
Ripple Counter 4 bits with JK Flip-Flop



Timing Diagram



1. [A] Ripple Counter



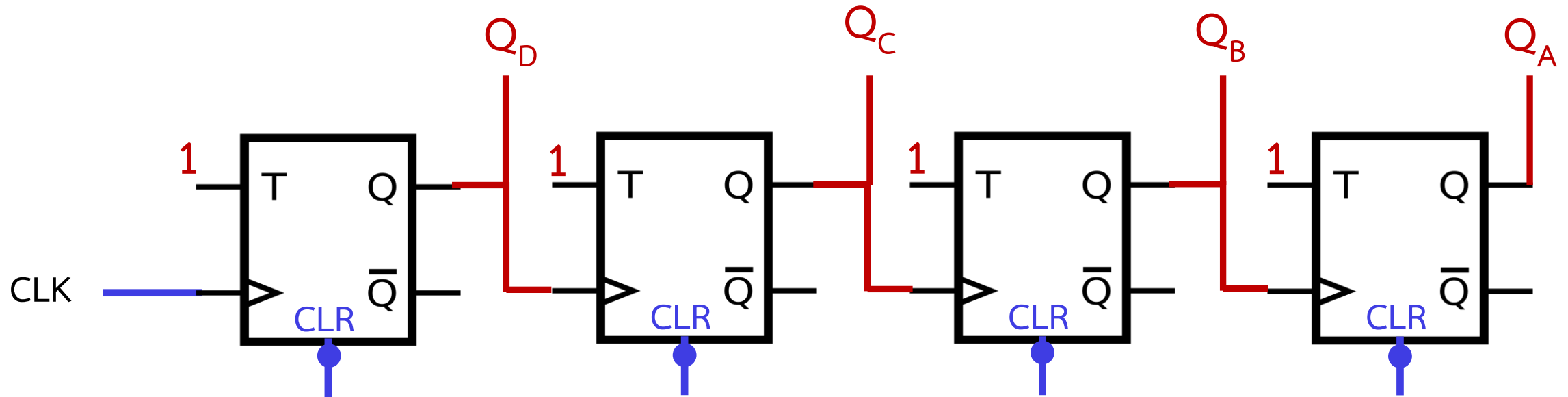
MSB : Flip-Flop [A]

LSB : Flip-Flop [D]

Counter Truth Table

Clock	A	B	C	D
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1
16	0	0	0	0
17	0	0	0	1

1. [A] Ripple Counter

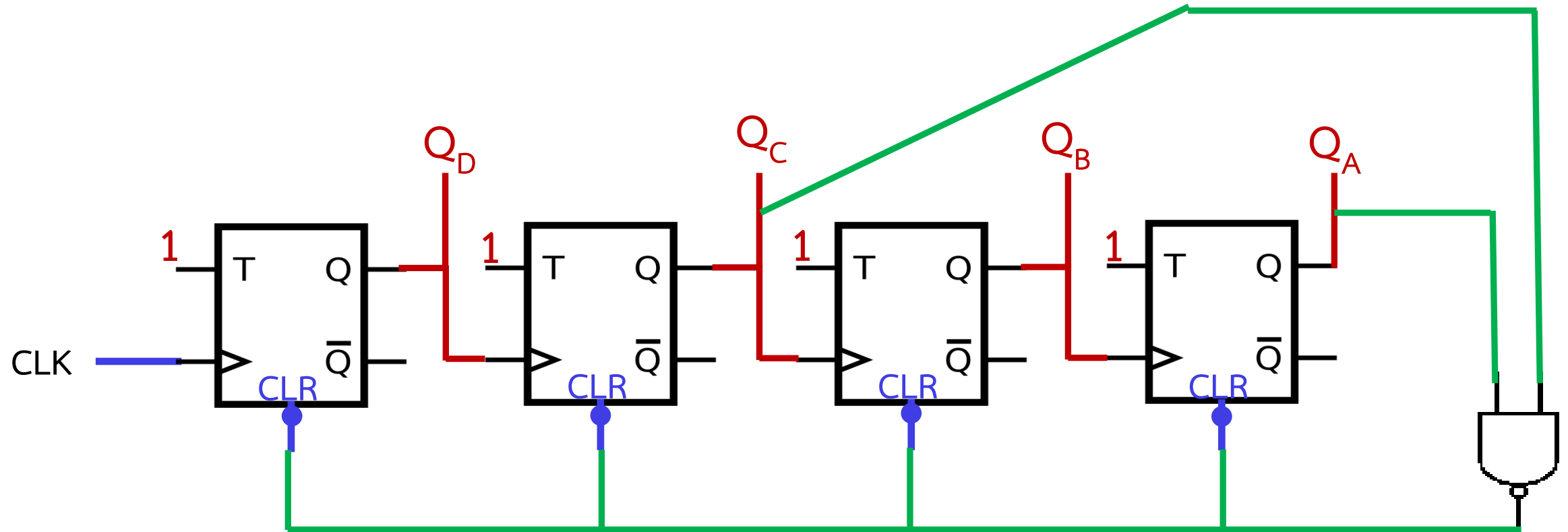


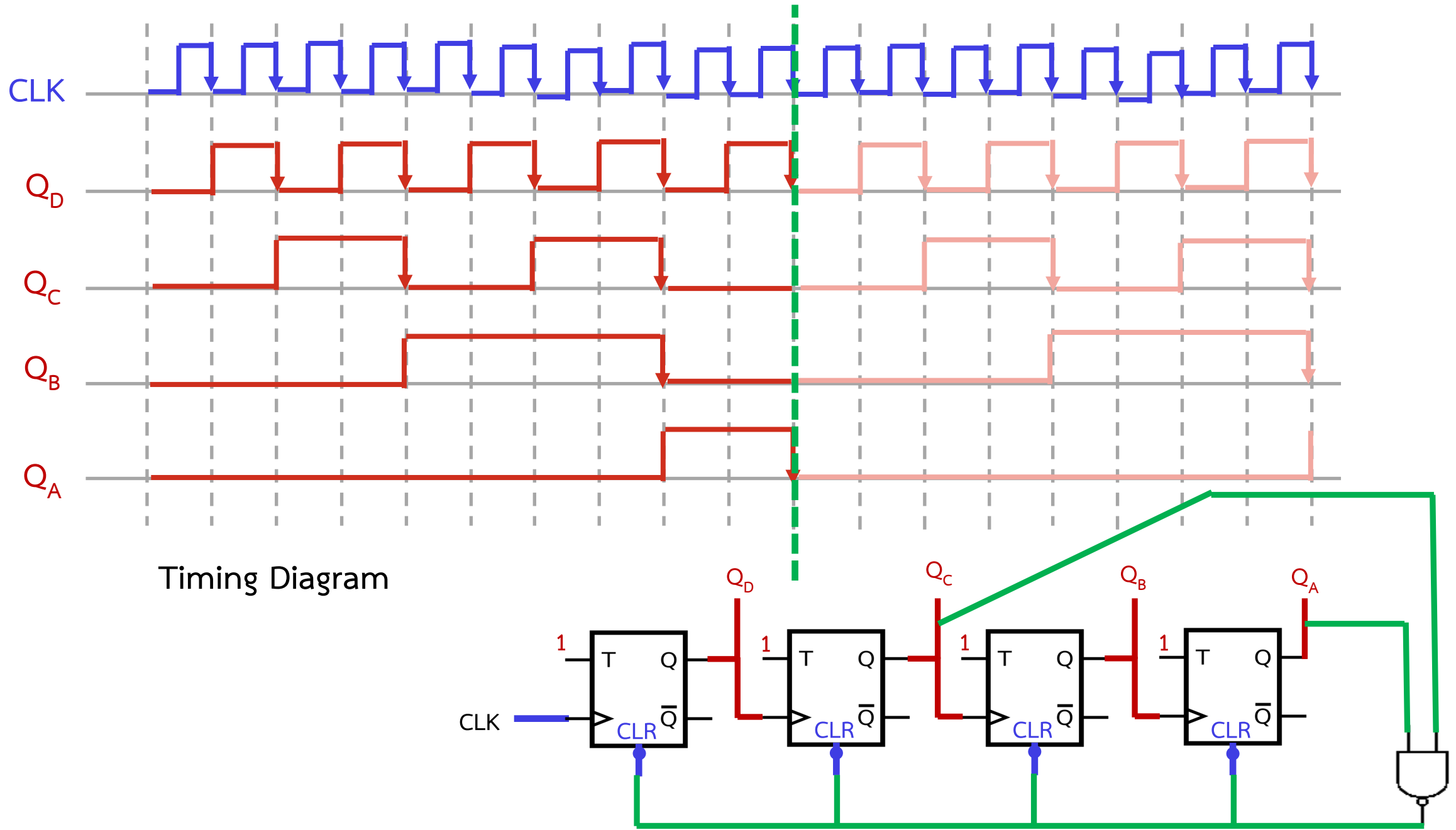
Ripple Counter 4 bits with T Flip-Flop

1. [A] Ripple Counter

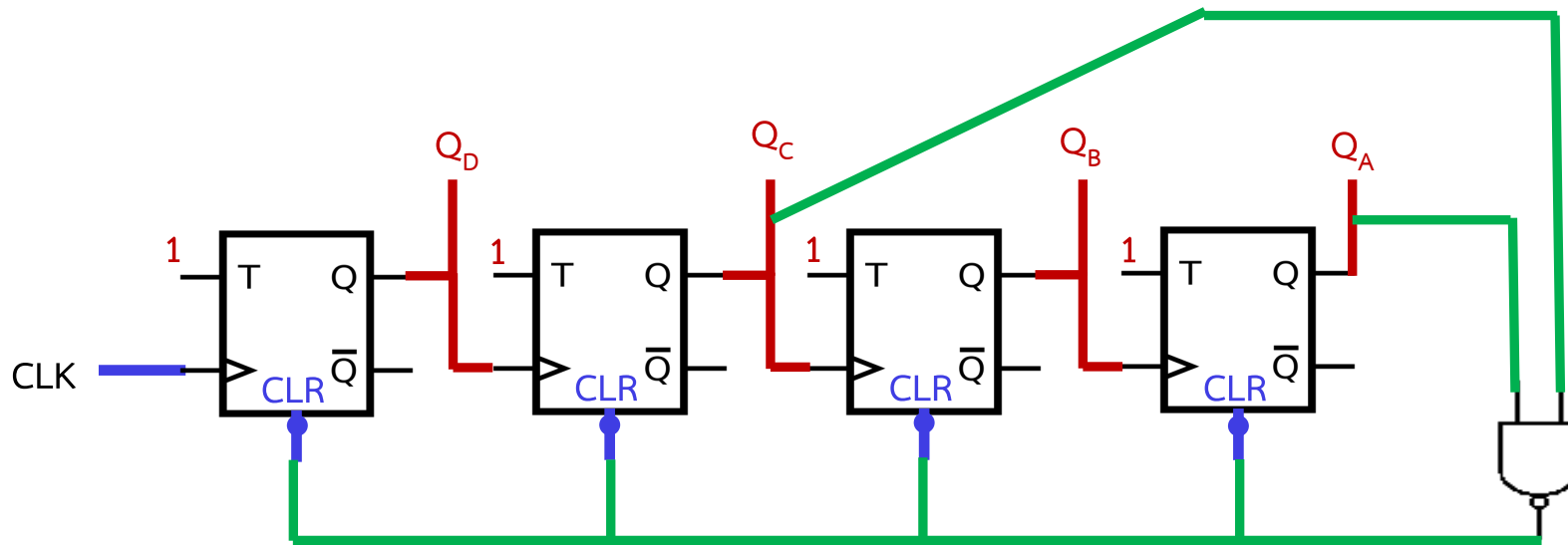


1. [B] Modulo-M Counter





1. [B] Modulo-M Counter



MSB : Flip-Flop [A]

LSB : Flip-Flop [D]

Counter Truth Table

Clock	A	B	C	D
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	0	0	0	0
11	0	0	0	1
12	0	0	1	0
13	0	0	1	1
14	0	1	0	0
15	0	1	0	1
16	0	1	1	0
17	0	1	1	1

1. [B] Modulo-M Counter

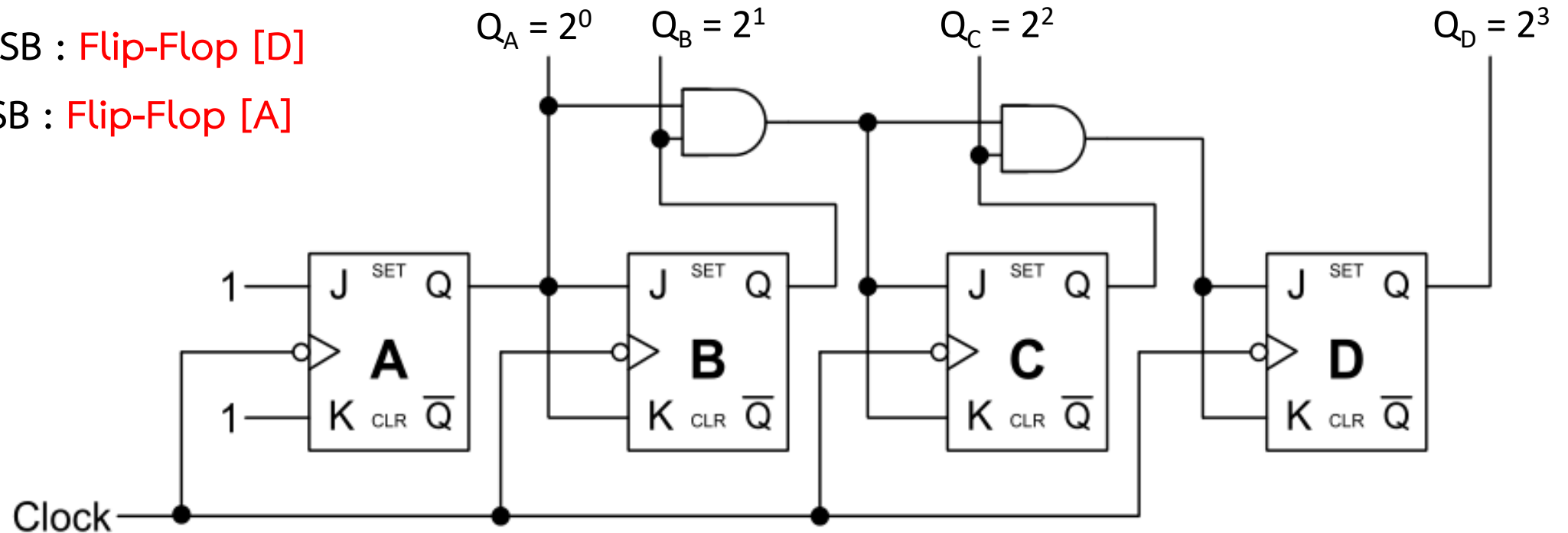


2. Synchronous Counter

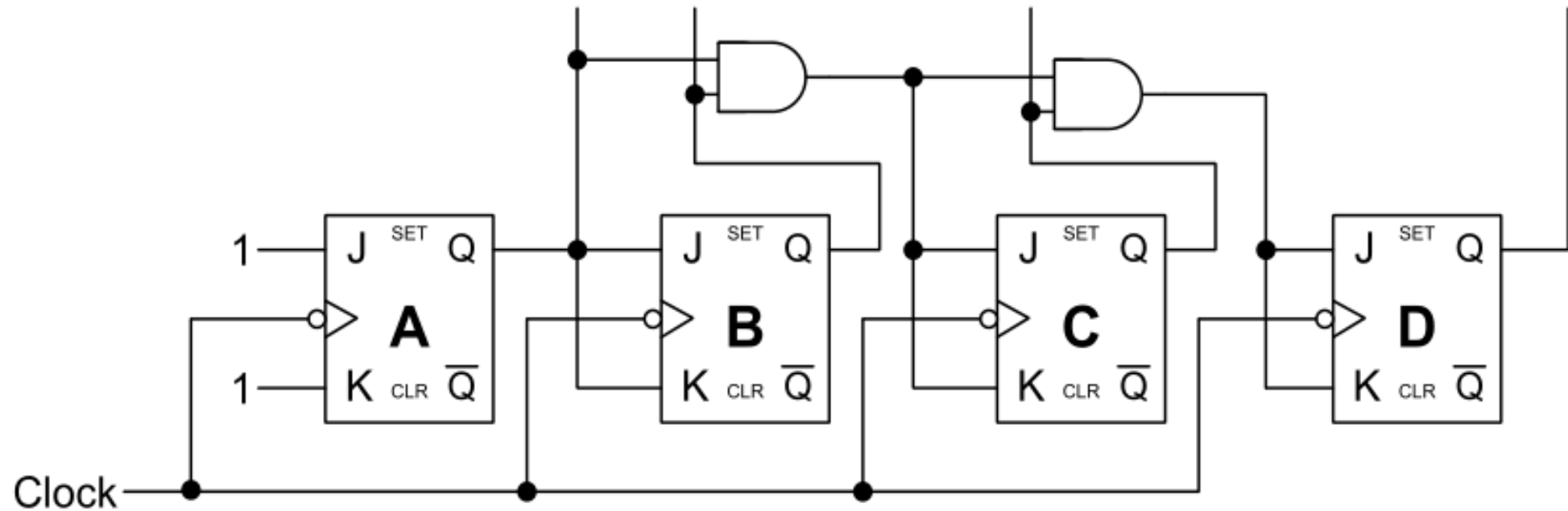
All flip-flops work at the same time. Clock pulse connect to all flip-flops.

MSB : Flip-Flop [D]

LSB : Flip-Flop [A]



2. Synchronous Counter



2. Synchronous Counter

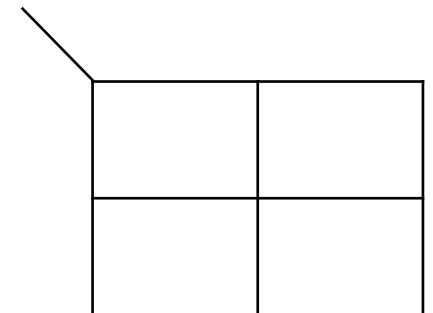
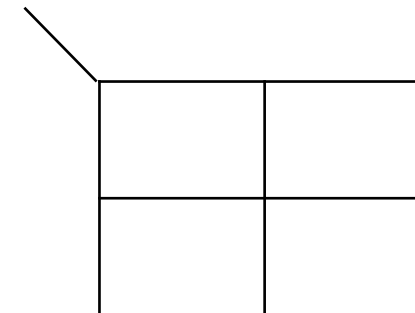
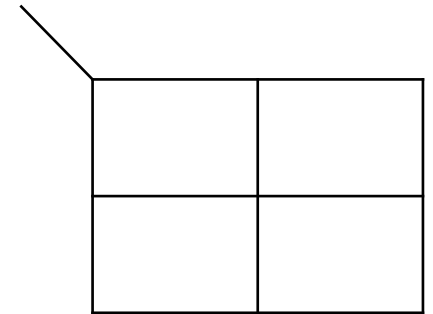
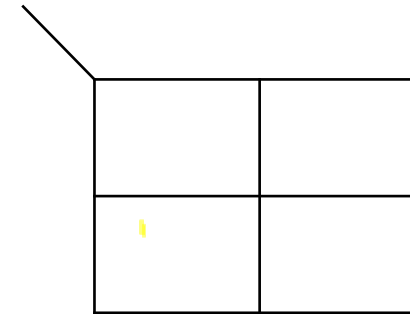


Circuit Design of Synchronous Counter

- 1.) Write an Excitation Table & Assign input for each input of flip-flop
- 2.) Create Boolean Algebra (Karnaugh-Map) for each input of flip-flop
- 3.) Draw a Circuit from Karnaugh-Map

Example #1

นับ Modulo 4 แบบ Synchronous โดยใช้ JK Flip-Flop



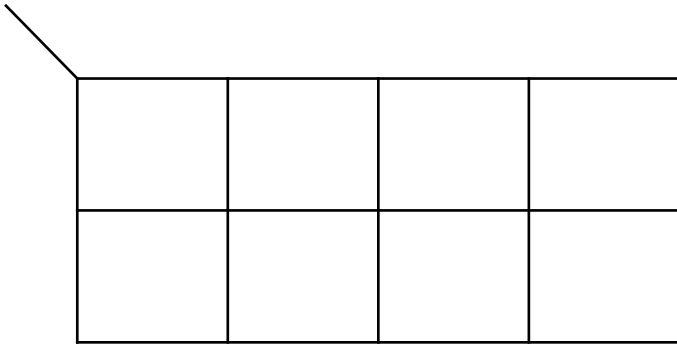
Example #1

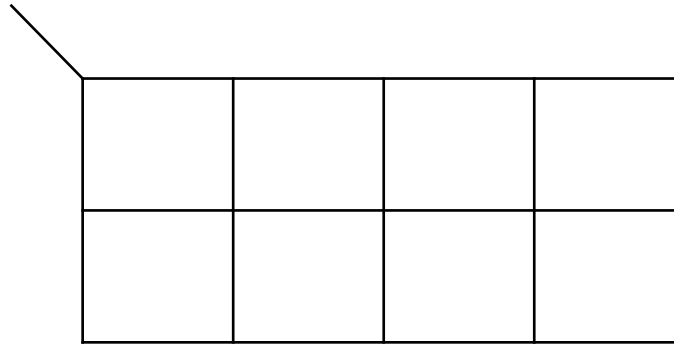
Example #2

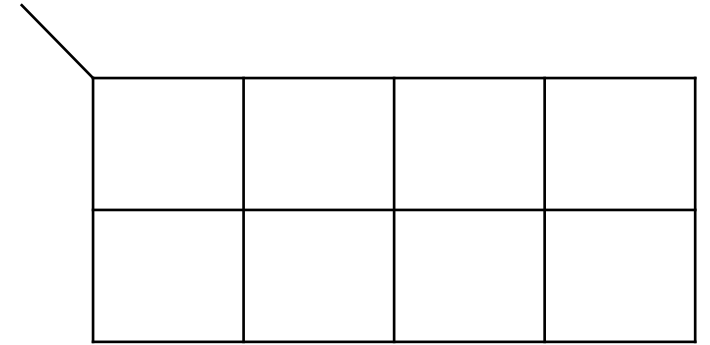
นับแบบกระโดด จาก 0 -> 6 -> 5 -> 2 -> 4 และวนกลับมาที่ 0 ใหม่ โดยใช้ JK Flip-Flop

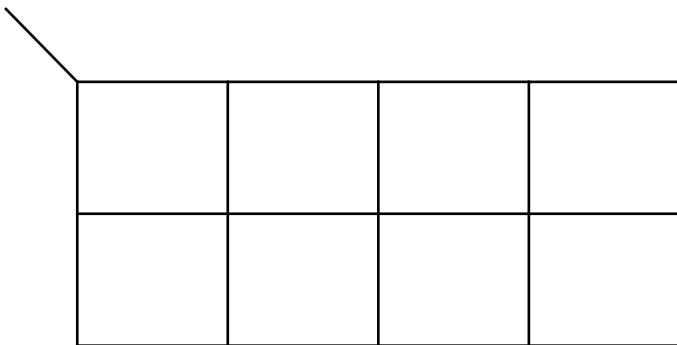
Example #2

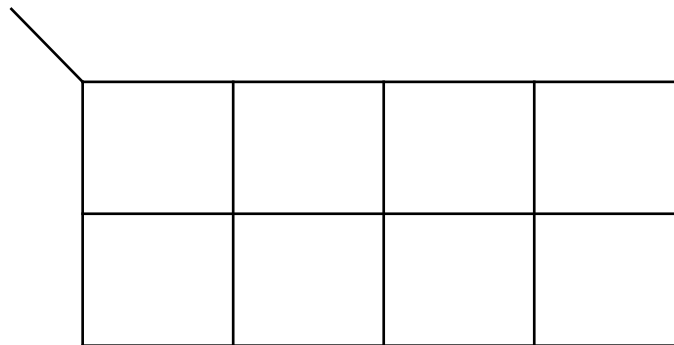
นับแบบกระโดด จาก 0 -> 6 -> 5 -> 2 -> 4

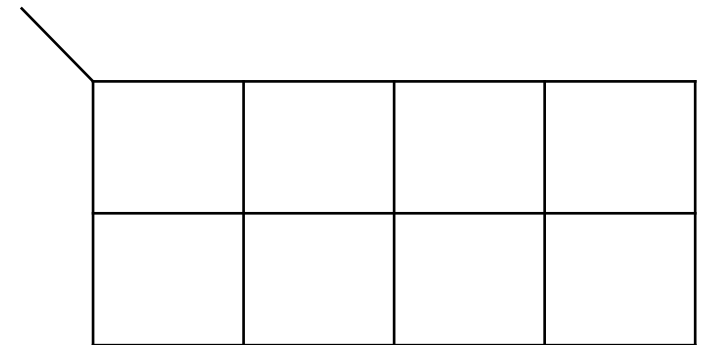












Example #2

3. Variable Modulus Counter

This design allows the circuit to change the last counted value.

4. Self Stopping Counter

This design allows the circuit to stop counting at the specific value.

